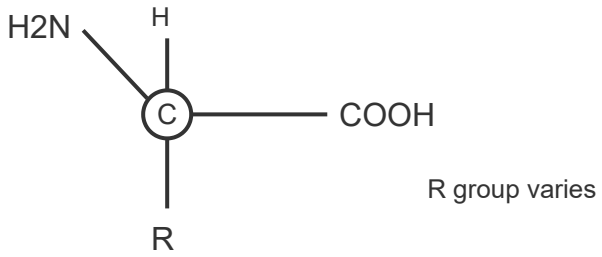
AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Proteins and DNA

Proteins do most of the work in a cell, and DNA holds the instructions for building them. This topic follows the story from amino acids, through the folding of a polypeptide into a working protein, to the double helix of DNA and the way it copies itself before each cell division.

1. Amino acids

Every amino acid has the same core. A central carbon atom carries an amine group (-NH2), a carboxyl group (-COOH), a hydrogen atom, and a variable side chain called the R group. The R group is what distinguishes one amino acid from another. Its size, charge and polarity decide how the amino acid behaves, and so how the finished protein folds.



2. Forming a polypeptide

Amino acids are joined by a peptide bond. The carboxyl group of one amino acid reacts with the amine group of the next, and a molecule of water is removed in a condensation reaction. The bond formed is the C-N link shown in the diagram. A chain of amino acids is a polypeptide, and a chain with two units is a dipeptide. Adding water back in breaks the peptide bond again, in a hydrolysis reaction.

The order of amino acids in the chain is the primary structure. That order is set by the sequence of bases in the DNA gene that codes for the protein, so DNA controls which protein is made.

3. How a protein folds

A chain of amino acids does not stay straight. It folds into a shape that is held in place by several kinds of bond, and each level of folding builds on the one below it.

- **Primary structure:** the sequence of amino acids joined by peptide bonds.
- **Secondary structure:** local folding of the chain into an alpha helix or a beta pleated sheet. Both are held together by hydrogen bonds between the C=O group of one amino acid and the N-H group of another, a few residues further along the chain.

- **Tertiary structure:** the overall three-dimensional shape of a single polypeptide, maintained by bonds between R groups. These are hydrogen bonds, ionic bonds between charged R groups, disulfide bridges between cysteine residues, and hydrophobic interactions.
- **Quaternary structure:** two or more polypeptide chains held together to form one functional protein.

The shape of the final protein is what matters. Hydrophilic R groups tend to sit on the outside of a soluble protein where they can meet water, and hydrophobic R groups tend to be buried in the middle. A protein's function depends on its shape, especially on the shape of any active site or binding site.

KEY IDEA

Primary structure is fixed by the gene. Secondary and tertiary structure are determined by the bonds that the amino acid sequence allows. Changing just one amino acid can change which R groups are close together, and that can change the whole shape.

4. Fibrous and globular proteins

Proteins fall into two broad shape types. Fibrous proteins are long and repetitive, with polypeptides lying parallel to one another. They are usually insoluble and strong, and they are good for structure. Collagen, found in tendons, ligaments, bone and skin, is a fibrous protein with three polypeptides wound into a triple helix. Repeated small amino acids such as glycine let the chains pack tightly, and many hydrogen bonds hold the helix together. This gives collagen high tensile strength.

Globular proteins are folded into compact, roughly spherical shapes. Their hydrophilic R groups face outwards, which makes them soluble in water. Their precise shape allows them to bind other molecules, so globular proteins act as enzymes, carriers and transport proteins. Haemoglobin is a globular protein made of four polypeptide chains, two alpha and two beta, each holding one haem group that contains an iron ion. Each haem group can bind one oxygen molecule, so one haemoglobin molecule carries four.

Feature	Fibrous protein (e.g. collagen)	Globular protein (e.g. haemoglobin)
Overall shape	Long and rod-like	Compact and roughly spherical
Solubility	Insoluble	Soluble
Amino acid pattern	Repetitive, often small R groups	Varied, with hydrophilic R groups outside
Main job	Structural support	Transport, binding and catalysis

5. Denaturation

Heat, extreme pH, high salt concentration and mechanical agitation all disturb the bonds that hold a protein in shape. The protein unfolds, its active or binding site is lost, and it stops working. Denaturation affects secondary, tertiary and quaternary structure, but the primary structure, the sequence of peptide bonds, is not broken. This is why a denatured protein is not destroyed in the chemical sense, but it no longer functions.

COMMON MISTAKE

Denaturation does not break peptide bonds. Breaking a peptide bond is hydrolysis, which needs water and an enzyme or strong acid, and it is a different process. Denaturation breaks the weaker bonds that shape the protein.

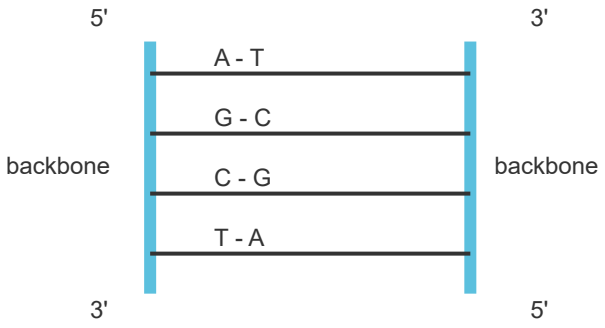
Enzymes are globular proteins, so temperature and pH change their rate of reaction by changing the shape of the active site.

6. The structure of DNA

DNA is a polymer of nucleotides. Each nucleotide has three parts: a phosphate group, a pentose sugar, and a nitrogen-containing base. In DNA the sugar is deoxyribose and the bases are adenine (A), thymine (T), cytosine (C) and guanine (G). Adenine and guanine are purines, which have a double-ring structure. Cytosine and thymine are pyrimidines, which have a single ring. RNA differs in having ribose instead of deoxyribose and uracil (U) in place of thymine.

Nucleotides link together by condensation reactions. The phosphate attached to one sugar joins to the third carbon of the sugar in the next nucleotide, forming a phosphodiester bond. This makes a sugar-phosphate backbone with the bases projecting from it. Each chain has a 5' end, which carries a phosphate, and a 3' end, which carries a free hydroxyl group.

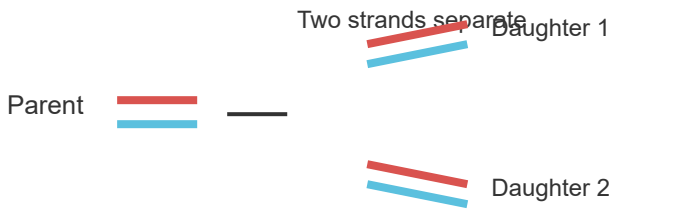
In DNA two chains run side by side in opposite directions, so they are antiparallel. The bases of one chain pair with those of the other by hydrogen bonds. Complementary pairing is strict: A pairs only with T, using two hydrogen bonds, and C pairs only with G, using three. Because every purine pairs with a pyrimidine, the two strands have the same width all the way along, and the whole molecule twists into a regular double helix with about ten base pairs per turn.



7. Semi-conservative DNA replication

Before a cell divides, its DNA must be copied so that each daughter cell receives a full set. Replication starts when the enzyme helicase breaks the hydrogen bonds between the base pairs, so the double helix unwinds and the two strands separate. Each strand now acts as a template. Free nucleotides in the nucleus pair with exposed bases by complementary base pairing: A with T, and C with G.

DNA polymerase then joins the neighbouring nucleotides along each new chain, forming phosphodiester bonds through condensation. The result is two daughter molecules. Each one contains one original strand and one newly made strand. This is why the process is called semi-conservative: half of each molecule is conserved from the parent.



Red = original strand, blue = new strand (each daughter keeps c

8. The Meselson and Stahl experiment

Three replication models were possible. In the conservative model the original double helix stays intact and a completely new copy is made. In the semi-conservative model each daughter keeps one old strand. In the dispersive model the old DNA is broken into pieces and mixed with new material, so every strand is part old and part new.

Meselson and Stahl tested these models by growing bacteria in medium containing the heavy isotope nitrogen-15, so their DNA became dense. They then moved the bacteria into medium with the light isotope nitrogen-14. DNA samples were spun in a density gradient, which separates molecules by density. Heavy DNA sits lowest, light DNA sits highest, and hybrid DNA, with one heavy and one light strand, sits between them.

Model	After one generation	After two generations
Conservative	Half heavy, half light (no intermediate band)	Heavy and light bands only
Semi-conservative	One intermediate band only	Intermediate and light bands in equal amounts
Dispersive	One intermediate band only	One intermediate band only

Compare the predictions of the three models generation by generation.

Showing that replication is semi-conservative

1. Grow the bacteria in heavy nitrogen for several generations so that all their DNA is heavy.
2. Transfer them to light nitrogen. After one generation there is only an intermediate band. The conservative model is ruled out, because it would give a heavy and a light band.
3. After a second generation there is an intermediate band and a light band. The dispersive model is ruled out, because it would give only intermediate DNA.

4. Conclude that each new molecule contains one original strand and one newly made strand, which is the semi-conservative model.

EXAM TIP

In an exam, state the band pattern for each model after each generation. Then say which model each result rules out. Simply naming the semi-conservative model is not enough.

9. Quick checks

QUICK CHECK

What type of bond joins two amino acids together in a polypeptide?

- Hydrogen bond
- Disulfide bridge
- Peptide bond
- Phosphodiester bond

Amino acids are linked by a peptide bond, formed by condensation between the carboxyl group of one and the amine group of the next. Hydrogen bonds and disulfide bridges hold the folded protein together rather than linking the chain, and phosphodiester bonds join nucleotides in DNA.

QUICK CHECK

Which pair of bases is joined by three hydrogen bonds in DNA?

- Adenine and thymine
- Adenine and guanine
- Thymine and cytosine
- Cytosine and guanine

Cytosine and guanine form three hydrogen bonds, while adenine and thymine form two. Adenine and guanine are both purines, so they cannot pair with each other, and thymine and cytosine are both pyrimidines.

QUICK CHECK

In the Meselson and Stahl experiment, what result after one generation in light nitrogen supports semi-conservative replication?

- Only a light band
- Equal heavy and light bands
- A single intermediate band
- Only a heavy band

Semi-conservative replication gives each new molecule one heavy and one light strand, which forms an intermediate band. A conservative model would have given half heavy and half light, and a light-only or heavy-only result would mean no replication had occurred.

QUICK CHECK

Which change would most directly denature a protein?

- Breaking the peptide bonds with water
- Adding a new amino acid to the end of the chain
- Heating it to a high temperature
- Increasing the length of the DNA gene

Heat disrupts the hydrogen, ionic and disulfide bonds that hold the folded shape, so the protein unfolds and loses function. Breaking peptide bonds is hydrolysis, not denaturation, and the other two options do not change the existing protein's shape.